APPENDIX II—GUINEA: JOINT BANK-FUND DEBT SUSTAINABILITY ANALYSIS UNDER THE DEBT SUSTAINABILITY FRAMEWORK FOR LOW-INCOME COUNTRIES[1]

The debt sustainability analysis (LIC-DSA) shows that Guinea is at a moderate risk of debt distress. After full HIPC, MDRI and beyond-HIPC debt relief, all external indicators remain under their indicative thresholds throughout the projection period. However, the country remains vulnerable to certain macroeconomic shocks, especially during the first half of the projection period. The public sector debt sustainability analysis (DSA) indicates that Guinea's domestic debt is significant but is expected to decrease over the longer run and does not alter the assessment. The remaining vulnerability to macroeconomic shocks indicates the need for prudent fiscal policies and debt management.

I. BACKGROUND

1. **This analysis (LIC-DSA) is based on the Debt Sustainability Framework (DSF) for Low-Income Countries**. The DSA presents the projected path of Guinea's external and public sector debt indicators, and assesses the country's risk of external debt distress. The LIC-DSA and the HIPC Initiative Debt Relief Analysis (HIPC-DRA) share the same macroeconomic assumptions for the baseline, but methodologically, they differ. The LIC-DSA compares the evolution over the projection period of debt burden indicators against policy-dependent indicative thresholds. In contrast, under the HIPC-DRA, the historical debt burden indicators are compared to uniform thresholds in order to calculate the amount of HIPC debt relief that Guinea qualifies for in the context of the HIPC Initiative. In addition, the results of the LIC-DSA differ from the HIPC-DRA because of methodological differences related to the definition of the discount rates and the exchange rates used.[2]

2. **This LIC-DSA[3] updates the analysis of the external and public debt of Guinea that was considered by the Executive Board of the IMF in February 2012.[4]** At that time Guinea had been in debt distress since 2007, as evidenced by the accumulation of external debt service arrears.

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[1] The DSA was prepared jointly by the staffs of the International Monetary Fund and the World Bank, in collaboration with the Authorities of Guinea. The fiscal year in Guinea is January 1 to December 31.

[2] The LIC-DSA and HIPC-DRA also differ because of the treatment of the French debt-for-development swaps (C2D).

[3] The DSAs presented in this document are based on the low-income countries (LIC) DSA framework. Under the Country Policy and Institutional Assessment (CPIA), Guinea is rated as a weak performer, with an average rating of 2.86 in 2008–10; the DSA uses the indicative threshold indicators for countries in this category. See "Debt Sustainability in Low-Income Countries: Proposal for an Operational Framework and Policy Implications" (http://www.imf.org/external/np/pdr/sustain/2004/020304.htm and IDA/SECM2004/0035, 2/3/04) and "Debt Sustainability in Low-Income Countries: Further Considerations on an Operational Framework, Policy Implications" http://www.imf.org/external/np/pdr/sustain/2004/091004.htm and IDA/SECM2004/0629, 9/10/04) and "A Review of Some Aspects of the Low-Income Country Debt Sustainability Framework" (http://www.imf.org/external/np/pp/eng/2009/080509a.pdf) and "Staff Guidance Note on the Application of the Joint Bank-Fund Debt Sustainability Framework for Low-Income Countries" (http://www.imf.org/external/np/pp/eng/2010/012210.pdf)

[4] See Supplement of IMF Country Report No. 12/63, March 2012, on LIC-DSA which can be found at: http://www.imf.org/external/pubs/ft/scr/2012/cr1263.pdf

3. **At end-2011, Guinea's public and publicly guaranteed external debt was $3,194 million, or 62 percent of 2011 GDP**.[5] The level of debt in nominal terms has been broadly stable in recent years, reflecting a low level of new loans and debt service paid, but with arrears accounting for a growing share (amounting to 12 percent of end-2011 debt outstanding).[6] Multilateral creditors accounted for 62 percent of the total, with the AfDB group and IDA accounting for almost four-fifths of the multilaterals' share. Paris Club creditors accounted for 25 percent of total, while official bilateral non-Paris Club and commercial creditors made up the rest.[7]

4. **External debt service arrears reached $371 million at end-2011**. Paris Club creditors accounted for 60 percent of the total. During 2011, the government cleared its arrears with all multilateral creditors and resumed debt service payments in full to them. HIPC interim assistance was discontinued by a majority of creditors in 2009–10.

5. **In April, 2012 the Paris Club and the Government of Guinea concluded a debt relief agreement on exceptional terms, including the rescheduling of post-cutoff date maturities and arrears**.[8] In addition, agreement was reached with European Union that outstanding arrears would in part be cancelled in 2011. Another part was frozen and will be cleared at the HIPC completion point using European Development Fund resources of the EU. As a result, outstanding arrears were reduced by about two-thirds; the remainder is to non-Paris Club official bilateral and commercial creditors on which no further debt service maturities fall due.

6. **The level of public domestic debt reflects the considerable increase in 2009–10 of advances by the central bank to the government**. During the military regime (2009–10), soaring expenditures were financed by borrowing from the domestic banking system. The liabilities of the government to the domestic banking system increased sharply from 13 percent of GDP in 2008, to 19 percent in 2009, and 30 percent in 2010. In 2011, large exceptional revenue from the mining sector (15 percent of GDP) was deposited at the central bank. The government has committed to refraining from any domestic bank financing of the budget other than stemming from the drawdown of this deposit. At end-June 2012, the stock of public domestic debt amounted to 11 percent of GDP, divided almost evenly between central bank advances and treasury bills held by domestic banks.

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[5] Excluding arrears to the EU which, by agreement, have been frozen pending clearance at the HIPC completion point using European Development Fund resources of the EU.

[6] This share includes arrears to Paris Club creditors which were restructured under the April 2012 debt relief agreement. Excluding these arrears the share of arrears in end-2011 debt would be 7 percent.

[7] Guinea has no debt service obligations falling due to commercial creditors, and arrears account for the full amount of debt outstanding.

[8] The cut-off date is January 1, 1986.

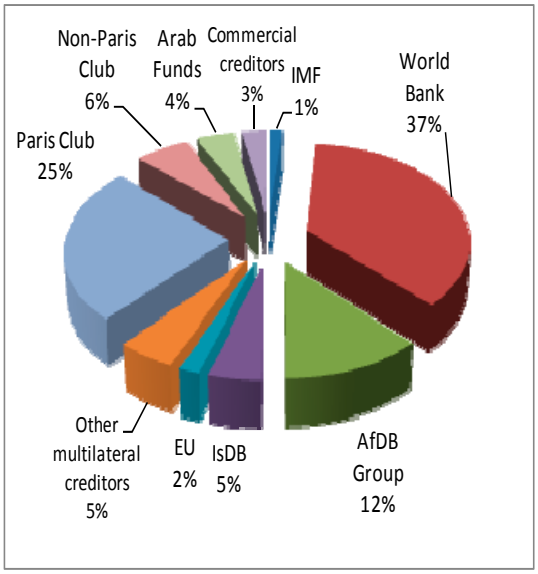
Guinea: Structure of External Public Debt (end-2011, nominal)

Text Table 1

	US$ million	Percent of GDP
Total	**3,194**	**61.77**
Multilateral creditors	**1,966**	**38.02**
IMF	42	0.80
World Bank	1,166	22.55
AfDB Group	393	7.60
IsDB	143	2.77
EU	59	1.14
Other multilateral creditors	163	3.15
Official bilateral creditors	**1,142**	**22.10**
Paris Club	813	15.73
Non-Paris Club	193	3.72
Arab Funds	137	2.65
Commercial creditors	**85**	**1.65**

Sources: Guinea authorities, and AfDB, World Bank, and IMF staff estimates

Text Figure 1



II. Baseline Assumptions

7. **The macroeconomic assumptions of the LIC-DSA are broadly consistent with those used in the HIPC-DRA and those of the February 2012 LIC-DSA** (Box 2 above). The baseline assumes a large increase in FDI which would underpin the start of large mining investments and a sizeable expansion in economic activity; political stability; sound macroeconomic management; prudent borrowing policies; and advancement in structural reforms over the medium term. It also assumes a substantial rise in public investment, especially in long-neglected infrastructure and the energy sector, as well as government support to develop agriculture and improve the business climate. These policies would provide a foundation for an increase in private investment, and would contribute to diversification of the economy and unlocking Guinea's long-run economic growth potential. Risks with regard to the macroeconomic projections include renewed political instability, especially in the run-up to parliamentary elections that were due in 2011 but that have been postponed several times, the potential further deterioration in the global economic outlook and mineral prices, and the possibility of projected mining production and revenues not materializing. With regard to external debt and financing needs, the DRA assumes all post-cut-off date ODA debt to be cancelled as part of beyond HIPC bilateral debt relief, and the LIC-DSA assumes that such claims held by France of €166 million (about $210 million) are to be converted into debt-for development-swaps (C2D) over 3 years.

8. **The baseline scenario reflects the full delivery of HIPC completion point, MDRI and beyond-HIPC relief, as in the HIPC-DRA**. **The results below are broadly in line with the alternative scenario related to HIPC and MDRI relief in the February 2012 LIC-DSA, with a number of differences**. The analysis discussed below incorporates the

impact of beyond HIPC bilateral relief. Also, updated projections of debt service (based on the HIPC completion point data reconciliation) and updated calculations of the delivery of HIPC and MDRI relief were used. These revisions lead to lower post-completion point debt burden indicators. In addition, the LIC-DSA assumes that French post-cut-off date ODA claims are converted into debt-for-development swaps (C2D) over 2012–15, thereby increasing debt service in the short term.[9] The initial fall in debt service relative to the 2012 LIC-DSA is dampened by the onset of the payments under C2D and the assumed restructuring of arrears to non-official bilateral and commercial creditors; under the 2012 Paris Club agreement payments to this group of creditors were already very low. Debt relief from multilateral financial institutions leads to a considerable reduction in debt service payments.

III. External Debt Sustainability Analysis

9. **Under the baseline scenario**, **Guinea's external debt indicators remain below their relevant indicative thresholds throughout the projection period** (Table 1, Figure 1). Debt stock and debt service ratios would immediately fall and remain below their policy thresholds. The sharp drop reflects in particular the impact of HIPC and MDRI relief from multilaterals, especially the World Bank, which holds the largest share of Guinea's multilateral debt (almost 60 percent at end-2011). The PV of external debt-to-GDP and PV of external debt-to-exports ratios fall in 2012 and again in 2015, reflecting the end of C2D payments, and remain relatively stable over the projection period. The stability of the ratios despite a rise in public investment relative to GDP reflects the impact of the projected sizeable expansion in the mining sector after 2015. This expansion not only leads to significant increases in exports and GDP, but also in fiscal revenues. In turn, the increase in fiscal revenues reduces the need for public borrowing to finance higher public investment. Debt service ratios also remain below their indicative thresholds over the projection period; the impact of the end of C2D payments is accentuated for these debt service indicators.

10. **Alternative scenarios and stress tests show that the external debt burden indicators are vulnerable to adverse shocks** (Figure 1):

- If the main economic variables remain at their historical level, and policy improvements and the expected growth dividend assumed under the baseline do not materialize, the PV of debt-to-GDP ratio would rise above the policy threshold after 2022, and then begin to decline toward the end of the projection period. The PV of debt-to-revenue also deteriorates, but remains below the policy threshold, and then declines over the longer run; the PV of debt-to-exports ratio shows a similar deterioration but continues to rise throughout the projection period while remaining below the policy threshold. The indicators are also very sensitive to exogenous shocks.

[9] Under C2D (*Contrats de désendettement et développement*) the existing debt service maturities are consolidated and re-profiled, and when they are paid, they are returned to the country through grants from France into development spending.

- Under most of the shocks considered in this analysis, the debt burden indicators would deteriorate;[10] the shocks on exports and on non-debt creating flows (such as net foreign direct investment), and a combination shock are particularly important resulting in a significant worsening of the debt burden indicators. This reflects the fact that the improvement in macroeconomic prospects depends heavily on the projected large inflow of foreign direct investments in the mining sector and the related jump in mining exports.

11. **Guinea's external debt sustainability will be affected by the modalities of the government's participation in large-scale infrastructure and mining projects**.[11] In the near-term this issue is raised in the context of the prospective SIMFER mining and infrastructure project. The total cost of the two projects is tentatively estimated at about $17–18 billion (one-third mining and two-thirds for infrastructure), of which the government has options for a substantial participation. Full take up of these options would entail commensurately large financial contributions to the projects. Moreover this issue would be raised in possible similar future projects. While no decisions have yet been taken by the government on how much of its possible stake to take up or on the financing of its contribution, it has stated clearly that its participation will not involve direct borrowing or guarantees by the government. Instead it intends, for these projects and possible similar future projects, to rely on private-public partnerships through special purpose financing vehicles for which financing costs would be fully covered by project revenues, and involve no contingent liabilities to the government. The participation of the IFC in the SIMFER and possible participation of the World Bank in the infrastructure project should help ensure that the projects are designed to be viable.

12. **While the use of PPPs to finance the government's participation in large-scale projects is a useful strategy to maintain external debt sustainability** , given the magnitude of these projects it will be important that the government strengthen its capacity to ensure that any financing agreement does not include contingent liabilities, and second that projections of revenues are well-based and sufficient to cover financing costs of the PPP and thus avoid possible implicit contingent liabilities in situations where the collapse of a project could have widespread damaging economic and social impacts. The assumption of any liability, could lead to sharp deterioration in debt burden indicators, particularly on debt service ratios.

IV. PUBLIC SECTOR DEBT SUSTAINABILITY

13. **The inclusion of domestic debt in the debt sustainability analysis worsens the debt burden indicators, although the domestic debt burden is expected to decrease over time (Table 3 and Figure 2)**. Following the large increase in borrowing from the domestic

[10] Simulations included shocks on GDP growth, export growth, inflation, non-debt creating flows, and exchange rate depreciation. The combination shock includes one-half standard deviation shocks to real GDP growth, exports, GDP deflator and net non-debt creating flows.

[11] See also discussion in the HIPC DRA.

banking system in 2009–10, the authorities virtually eliminated new borrowing in 2011, and in 2012–13 there is no new domestic financing planned for the budget, while net repayments of domestic debt are expected to resume.[12] As a result, the PV of public debt-to-GDP ratio would gradually decline from 29 percent of 2012 GDP to 9.8 percent of GDP at the end of the projection period. Debt service on total public debt would drop in 2012, and then again in 2015, and thereafter remain broadly stable reflecting largely the external debt service trend described above.

14. **The public debt position is vulnerable to shocks, particularly to policy reversals** (Table 4 and Figure 2). Under the fixed primary balance, historical average and the most extreme shock scenarios, the public debt burden indicators would be at least twice as high compared to the trajectories under the baseline scenario in the long term; in addition, the indicators show a rising trend over the projection period for the fixed primary balance scenario.

15. **In light of the results of the baseline and alternative scenarios as well as the stress tests, IDA and IMF staffs conclude that Guinea is at a moderate risk of debt distress**.

V. Conclusions

16. **This LIC-DSA shows that Guinea would be at a moderate risk of debt distress after the HIPC completion point**. This compares to the "in debt distress" assessment in the February 2012 LIC-DSA. Under the current baseline scenario, the HIPC Initiative, MDRI and beyond-HIPC relief significantly improve the external debt indicators over the projection period. All debt burden indicators remain below their policy-dependent thresholds throughout the projection period. Alternative scenarios and stress tests reveal the vulnerability of Guinea's external debt outlook, as the debt burden indicators rise and some breach the policy thresholds. The inclusion of domestic debt moderately weakens the debt outlook, but does not alter the assessment of Guinea's risk of debt distress.

17. **This LIC-DSA underscores the importance of sustained implementation of sound macroeconomic policies, improvements in the business environment, and prudent debt management**. Such policies are key to realizing the expected growth dividend, especially expansion of activity in the mining sector, and improved export and fiscal revenue performance, and with sound debt management are important for maintaining debt sustainability. In addition, the stress tests underscore the need for reducing the vulnerability of revenue and exports to commodity prices, in particular of metals, through output and export diversification. Debt relief under the HIPC Initiative, MDRI and beyond-HIPC assistance provides significant space for Guinea to access more concessional financing, increasing its ability to address its considerable public investment needs and improving growth prospects. However, in light of remaining vulnerabilities, large-scale new borrowing

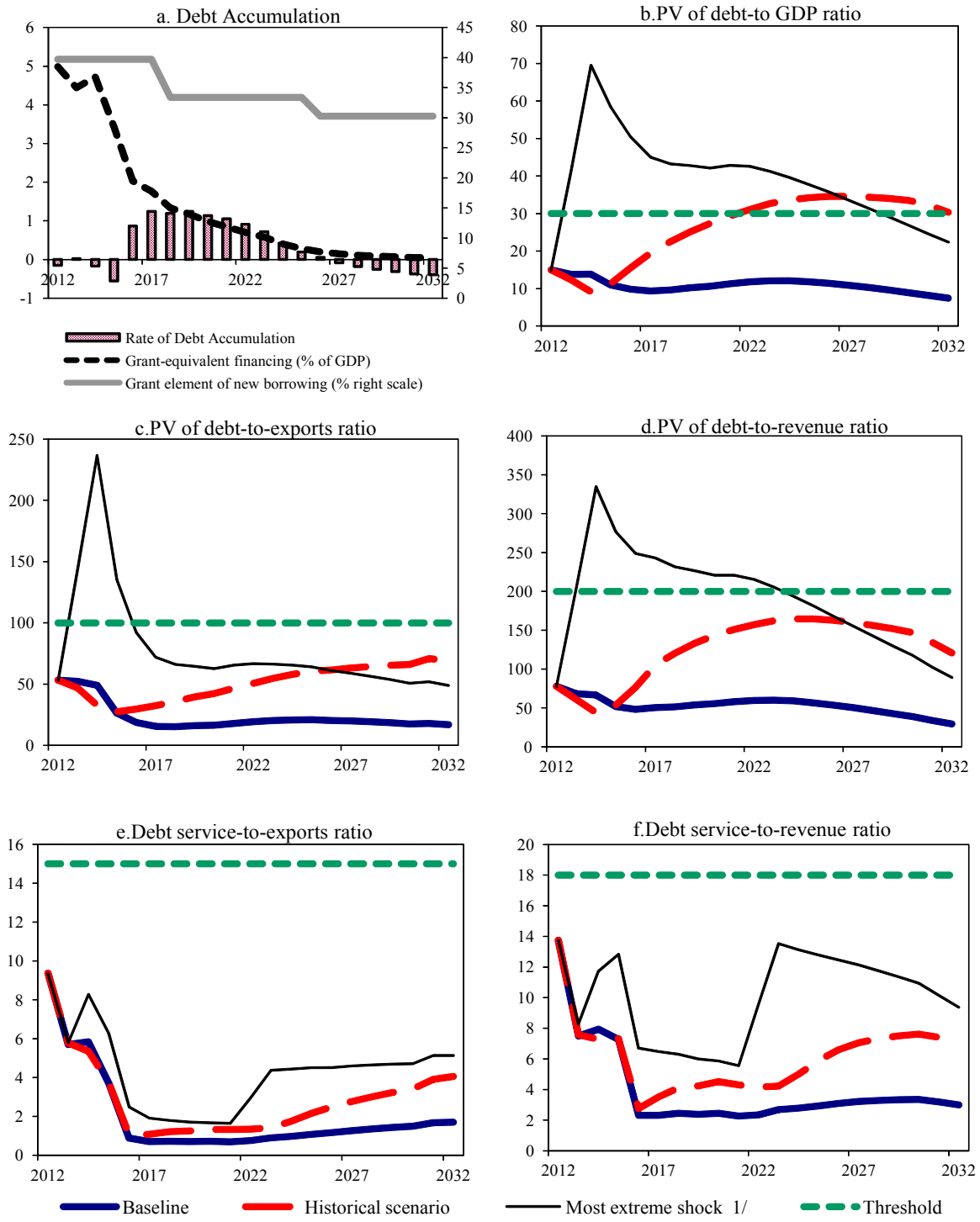
[12] Under an agreement between the Ministry of Economy and Finance and the central bank, the government will repay its advances over a period of 40 years, starting in 2020.

should be avoided and the authorities have emphasized their commitment to ensure that the government's participation in large-scale infrastructure projects and mining projects does not involve the use of budget resources to meet the associated financing needs; instead the government intends to make use of public-private partnerships (PPPs) in which project revenues are used to service debt committed under the project, thus avoiding significant government borrowing on nonconcessional terms.[13] However, to ensure that PPP agreements do not entail contingent liabilities, the administrative capacity to evaluate such liabilities needs to be strengthened and a clear regulatory framework for such partnerships needs to be established.

18 **The Guinean authorities broadly concurred with the assumptions and conclusions of the DSA**.

[13] See Supplement of IMF Country Report No. 12/63, March 2012 on LIC-DSA for a fuller discussion of this issue.

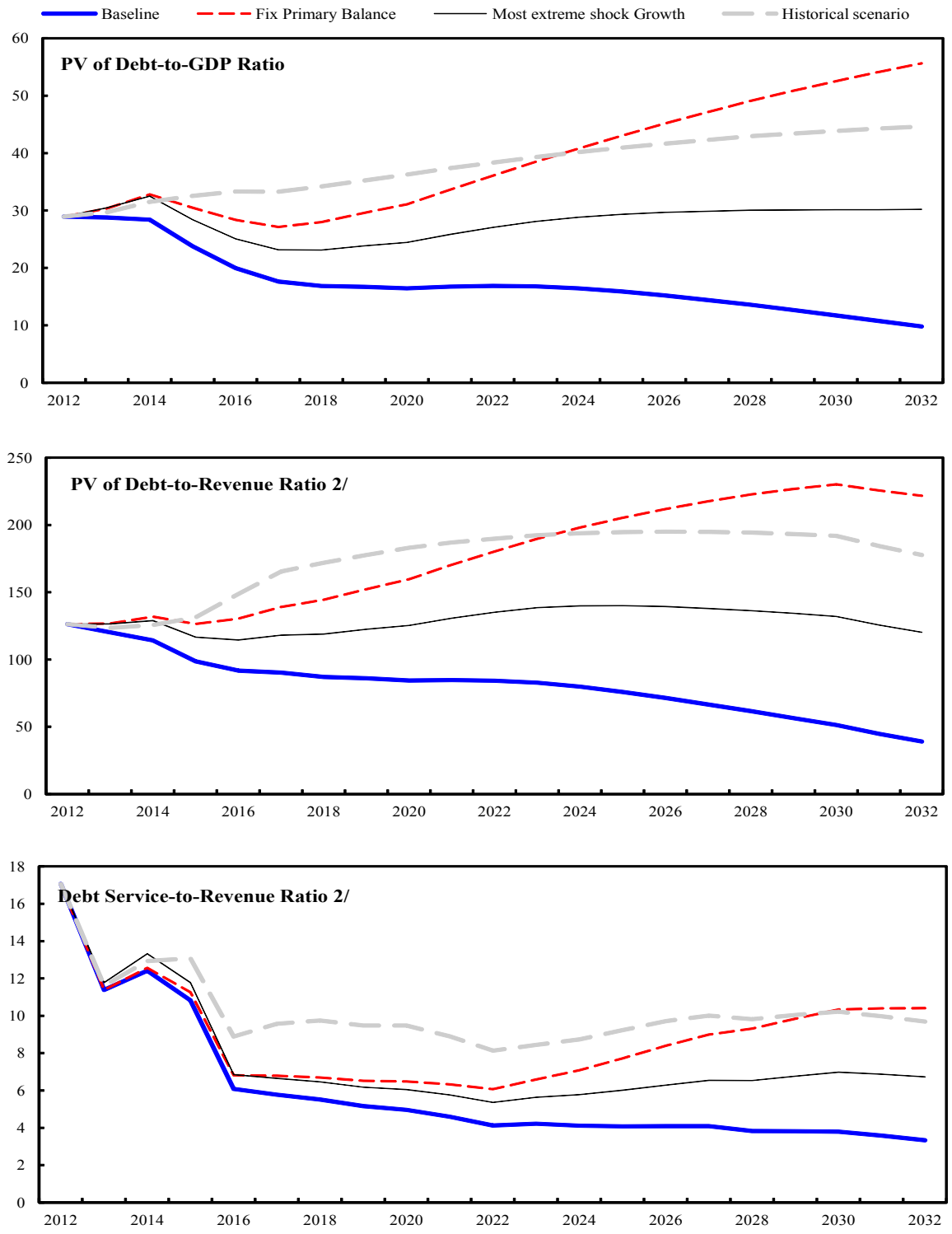
Figure 1. Guinea: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2012-2032 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2022. In figure b. it corresponds to a Combination shock; in c. to a Combination shock; in d. to a Combination shock; in e. to a Combination shock and in figure f. to a Combination shock

Figure 2.Guinea: Indicators of Public Debt Under Alternative Scenarios, 2012-2032 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2022.
2/ Revenues are defined inclusive of grants.

Table 1.: External Debt Sustainability Framework, Baseline Scenario, 2009-2032 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2009	2010	2011			2012	2013	2014	2015	2016	2017	2012-2017 Average	2022	2032	2018-2032 Average
External debt (nominal) 1/	**71.5**	**70.5**	**65.9**			**19.0**	**17.9**	**18.3**	**14.8**	**13.4**	**13.0**		**16.3**	**9.7**	
o/w public and publicly guaranteed (PPG)	71.5	70.5	65.9			19.0	17.9	18.3	14.8	13.4	13.0		16.3	9.7	
Change in external debt	-6.0	-1.0	-4.5			-46.9	-1.2	0.4	-3.5	-1.4	-0.4		0.6	-1.0	
Identified net debt-creating flows	4.9	5.7	3.7			7.3	3.0	4.0	-6.1	-6.8	-5.3		-1.9	0.6	
Non-interest current account deficit	**9.2**	**11.6**	**15.6**	**5.9**	**5.9**	**38.4**	**39.5**	**40.6**	**22.4**	**5.0**	**-0.4**		**2.1**	**5.5**	2.7
Deficit in balance of goods and services	4.3	8.2	17.9			36.2	37.0	36.9	19.7	2.4	-12.9		-11.0	-3.6	
Exports	26.5	28.4	30.3			28.1	26.4	28.2	41.4	52.6	60.2		61.3	43.9	
Imports	30.8	36.5	48.2			64.3	63.5	65.2	61.1	55.0	47.3		50.3	40.4	
Net current transfers (negative = inflow)	-5.3	-5.1	-10.4	-6.3	2.7	-5.9	-5.6	-4.6	-4.4	-3.7	-3.2		-2.5	-1.9	-2.4
o/w official	0.0	0.0	-2.2			-0.8	-1.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other current account flows (negative = net inflow)	10.3	8.6	8.1			8.1	8.1	8.2	7.1	6.2	15.7		15.7	11.0	
Net FDI (negative = inflow)	**-3.0**	**-2.4**	**-9.5**	**-4.3**	**3.0**	**-28.9**	**-35.8**	**-35.9**	**-25.7**	**-9.6**	**-3.5**		**-3.8**	**-4.7**	-4.1
Endogenous debt dynamics 2/	**-1.3**	**-3.5**	**-2.4**			**-2.3**	**-0.7**	**-0.7**	**-2.8**	**-2.2**	**-1.4**		**-0.2**	**-0.2**	
Contribution from nominal interest rate	0.7	0.7	0.9			0.6	0.2	0.2	0.2	0.2	0.2		0.2	0.1	
Contribution from real GDP growth	0.2	-1.3	-2.6			-2.8	-0.9	-0.9	-3.0	-2.4	-1.6		-0.4	-0.3	
Contribution from price and exchange rate changes	-2.2	-3.0	-0.7			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**-10.9**	**-6.7**	**-8.2**			**-54.2**	**-4.2**	**-3.6**	**2.6**	**5.4**	**5.0**		**2.5**	**-1.6**	
o/w exceptional financing	0.0	-0.1	-1.3			-37.5	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	…	…	37.1			14.9	13.8	13.8	11.0	9.8	9.4		11.8	7.4	
In percent of exports	…	…	122.4			53.2	52.2	49.0	26.5	18.6	15.6		19.3	16.9	
PV of PPG external debt	**…**	**…**	**37.1**			**14.9**	**13.8**	**13.8**	**11.0**	**9.8**	**9.4**		**11.8**	**7.4**	
In percent of exports	**…**	**…**	**122.4**			**53.2**	**52.2**	**49.0**	**26.5**	**18.6**	**15.6**		**19.3**	**16.9**	
In percent of government revenues	**…**	**…**	**220.8**			**78.0**	**68.5**	**66.7**	**51.9**	**48.3**	**50.6**		**59.7**	**29.6**	
Debt service-to-exports ratio (in percent)	**9.8**	**10.3**	**14.0**			**9.4**	**5.7**	**5.8**	**3.7**	**0.9**	**0.7**		**0.8**	**1.7**	
PPG debt service-to-exports ratio (in percent)	**9.8**	**10.3**	**14.0**			**9.4**	**5.7**	**5.8**	**3.7**	**0.9**	**0.7**		**0.8**	**1.7**	
PPG debt service-to-revenue ratio (in percent)	**16.1**	**19.2**	**25.3**			**13.7**	**7.5**	**7.9**	**7.3**	**2.3**	**2.3**		**2.3**	**3.0**	
Total gross financing need (Billions of U.S. dollars)	0.4	0.6	0.5			0.7	0.3	0.4	-0.1	-0.4	-0.4		-0.2	0.3	
Non-interest current account deficit that stabilizes debt ratio	15.2	12.7	20.1			85.3	40.7	40.2	25.9	6.4	0.0		1.5	6.5	
Key macroeconomic assumptions															
Real GDP growth (in percent)	-0.3	1.9	3.9	2.7	1.7	4.8	5.0	5.2	19.9	19.7	14.3	11.5	2.6	3.1	3.6
GDP deflator in US dollar terms (change in percent)	2.9	4.3	0.9	4.5	15.8	6.0	3.4	-5.8	0.9	0.9	3.3	1.4	0.9	1.4	1.1
Effective interest rate (percent) 5/	0.9	1.1	1.3	1.4	0.3	0.9	1.0	1.2	1.4	1.5	1.6	1.3	1.6	1.4	1.5
Growth of exports of G&S (US dollar terms, in percent)	-22.1	13.6	12.1	7.8	14.4	3.0	2.2	5.8	77.5	53.3	35.1	29.5	0.9	1.0	2.7
Growth of imports of G&S (US dollar terms, in percent)	-21.2	26.2	38.3	12.5	16.6	48.3	7.2	1.7	13.4	8.8	1.4	13.5	1.5	4.3	3.8
Grant element of new public sector borrowing (in percent)	…	…	…	…	…	39.7	39.7	39.7	39.7	39.7	39.7	39.7	33.4	30.3	31.9
Government revenues (excluding grants, in percent of GDP)	16.2	15.3	16.8			19.2	20.2	20.8	21.1	20.3	18.5		19.8	25.1	21.1
Aid flows (in Billions of US dollars) 7/	0.0	0.0	0.2			0.3	0.3	0.3	0.3	0.2	0.2		0.1	0.0	
o/w Grants	0.0	0.0	0.2			0.2	0.2	0.3	0.2	0.1	0.1		0.0	0.0	
o/w Concessional loans	0.0	0.0	0.0			0.1	0.1	0.1	0.0	0.1	0.1		0.1	0.0	
Grant-equivalent financing (in percent of GDP) 8/	…	…	…			5.0	4.4	4.7	3.4	2.0	1.8		0.7	0.0	0.5
Grant-equivalent financing (in percent of external financing) 8/	…	…	…			72.6	81.8	83.6	84.5	70.9	60.4		43.4	41.2	44.6
Memorandum items:															
Nominal GDP (Billions of US dollars)	4.6	4.9	5.2			5.7	6.2	6.2	7.5	9.0	10.7		14.3	21.4	
Nominal dollar GDP growth	2.6	6.3	4.9			11.1	8.6	-0.9	21.0	20.8	18.1	13.1	3.5	4.5	4.8
PV of PPG external debt (in Billions of US dollars)			1.8			0.8	0.8	0.8	0.8	0.9	1.0		1.7	1.6	
(PVt-PVt-1)/GDPt-1 (in percent)						-0.2	0.0	-0.2	-0.6	0.9	1.2	0.2	0.9	-0.4	0.4
Gross workers' remittances (Billions of US dollars)	…	…	…			…	…	…	…	…	…		…	…	
PV of PPG external debt (in percent of GDP + remittances)	…	…	37.1			14.9	13.8	13.8	11.0	9.8	9.4		11.8	7.4	
PV of PPG external debt (in percent of exports + remittances)	…	…	122.4			53.2	52.2	49.0	26.5	18.6	15.6		19.3	16.9	
Debt service of PPG external debt (in percent of exports + remittances)	…	…	14.0			9.4	5.7	5.8	3.7	0.9	0.7		0.8	1.7	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2.Guinea: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2012-2032
(In percent)

	Projections							
	2012	2013	2014	2015	2016	2017	**2022**	2032
PV of debt-to GDP ratio								
Baseline	15	14	14	11	10	9	**12**	7
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 1/	15	12	9	11	16	19	**31**	30
A2. New public sector loans on less favorable terms in 2012-2032 2	15	14	15	12	11	11	**16**	12
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	15	14	15	12	10	10	**13**	8
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	15	15	19	15	14	13	**15**	9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	15	16	17	13	12	11	**14**	9
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	15	39	64	54	46	41	**39**	20
B5. Combination of B1-B4 using one-half standard deviation shocks	15	41	70	58	50	45	**43**	22
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	15	19	19	15	14	13	**16**	10
PV of debt-to-exports ratio								
Baseline	53	52	49	26	19	16	**19**	17
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 1/	53	47	33	27	30	32	**51**	69
A2. New public sector loans on less favorable terms in 2012-2032 2	53	53	52	29	21	18	**25**	28
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	53	51	48	26	18	15	**19**	17
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	53	63	83	46	32	26	**29**	25
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	53	51	48	26	18	15	**19**	17
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	53	146	226	130	88	69	**64**	47
B5. Combination of B1-B4 using one-half standard deviation shocks	53	144	237	136	92	72	**67**	49
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	53	51	48	26	18	15	**19**	17
PV of debt-to-revenue ratio								
Baseline	78	68	67	52	48	51	**60**	30
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 1/	78	61	44	54	77	105	**157**	121
A2. New public sector loans on less favorable terms in 2012-2032 2	78	70	70	56	54	60	**78**	48
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	78	70	71	55	51	54	**63**	31
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	78	75	91	73	67	68	**73**	35
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	78	79	81	63	59	61	**72**	36
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	78	191	308	255	229	223	**197**	82
B5. Combination of B1-B4 using one-half standard deviation shocks	78	205	335	277	249	243	**215**	89
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	78	95	92	72	67	70	**82**	41

Table 2.Guinea: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2012-2032 (continued)
(In percent)

Debt service-to-exports ratio								
Baseline	9	6	6	4	1	1	**1**	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 1/	9	6	5	4	1	1	**1**	4
A2. New public sector loans on less favorable terms in 2012-2032 2	9	6	6	4	1	1	**1**	2
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	9	6	6	4	1	1	**1**	2
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	9	6	7	5	1	1	**1**	3
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	9	6	6	4	1	1	**1**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	9	6	8	6	2	2	**3**	5
B5. Combination of B1-B4 using one-half standard deviation shocks	9	6	8	6	2	2	**3**	5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	9	6	6	4	1	1	**1**	2
Debt service-to-revenue ratio								
Baseline	14	7	8	7	2	2	**2**	3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 1/	14	8	7	7	3	4	**4**	7
A2. New public sector loans on less favorable terms in 2012-2032 2	14	7	8	7	2	3	**3**	4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	14	8	9	8	3	3	**3**	3
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	14	7	8	8	3	3	**3**	4
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	14	9	10	9	3	3	**3**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	14	7	11	12	6	6	**9**	9
B5. Combination of B1-B4 using one-half standard deviation shocks	14	8	12	13	7	6	**10**	9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	14	11	11	10	3	3	**3**	4
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	32	32	32	32	32	32	**32**	32

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3.Guinea: Public Sector Debt Sustainability Framework, Baseline Scenario, 2009-2032
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2009	2010	2011	Average 5/	Standard Deviation 5/	2012	2013	2014	2015	2016	2017	2012-17 Average	2022	2032	2018-32 Average
Public sector debt 1/	90.1	100.7	76.8			33.0	32.9	32.8	27.6	23.5	21.3		21.4	12.1	
o/w foreign-currency denominated	71.5	70.5	65.9			19.0	17.9	18.3	14.8	13.4	13.0		16.3	9.7	
Change in public sector debt	0.2	10.6	-23.9			-43.8	-0.2	0.0	-5.3	-4.1	-2.3		0.2	-1.2	
Identified debt-creating flows	-1.7	11.1	-11.4			-43.5	-0.3	-0.4	-5.5	-4.3	-2.4		0.0	-1.2	
Primary deficit	5.0	12.0	-0.6	2.0	4.4	4.0	1.6	-0.2	-0.4	0.0	0.8	1.0	0.5	-0.8	-0.1
Revenue and grants	16.5	15.7	20.3			22.9	23.9	24.9	24.1	21.8	19.5		20.0	25.1	
of which: grants	0.4	0.4	3.4			3.7	3.8	4.1	3.0	1.5	1.0		0.2	0.0	
Primary (noninterest) expenditure	21.6	27.7	19.6			26.9	25.6	24.7	23.7	21.8	20.3		20.5	24.3	
Automatic debt dynamics	-6.7	-0.8	-9.5			-10.1	-1.9	-0.2	-5.1	-4.3	-3.2		-0.5	-0.4	
Contribution from interest rate/growth differential	0.8	-3.5	-8.0			-4.2	-1.6	-1.6	-5.2	-4.4	-3.0		-0.6	-0.5	
of which: contribution from average real interest rate	0.6	-1.8	-4.2			-0.7	-0.1	0.0	0.3	0.2	-0.1		-0.1	-0.1	
of which: contribution from real GDP growth	0.3	-1.7	-3.8			-3.5	-1.6	-1.6	-5.4	-4.5	-2.9		-0.5	-0.4	
Contribution from real exchange rate depreciation	-7.5	2.7	-1.5			-5.8	-0.3	1.4	0.1	0.1	-0.2		...	...	
Other identified debt-creating flows	0.0	-0.1	-1.3			-37.5	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	-0.1	-1.3			-37.5	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	1.9	-0.5	-12.4			-0.3	0.1	0.3	0.3	0.2	0.2		0.2	0.0	
Other Sustainability Indicators															
PV of public sector debt	...	...	48.0			28.9	28.8	28.4	23.8	20.0	17.6		16.8	9.8	
o/w foreign-currency denominated	...	...	37.1			14.9	13.8	13.8	11.0	9.8	9.4		11.8	7.4	
o/w external	...	...	37.1			14.9	13.8	13.8	11.0	9.8	9.4		11.8	7.4	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	9.3	16.7	4.7			7.9	4.4	2.9	2.2	1.3	1.9		1.3	0.1	
PV of public sector debt-to-revenue and grants ratio (in percent)	…	…	237.0			126.2	120.2	114.2	98.6	91.8	90.4		84.2	39.1	
PV of public sector debt-to-revenue ratio (in percent)	…	…	285.7			150.9	142.9	136.9	112.6	98.5	95.3		85.2	39.1	
o/w external 3/	…	…	220.8			78.0	68.5	66.7	51.9	48.3	50.6		59.7	29.6	
Debt service-to-revenue and grants ratio (in percent) 4/	25.7	30.0	26.5			17.1	11.4	12.4	10.8	6.1	5.8		4.1	3.3	
Debt service-to-revenue ratio (in percent) 4/	26.3	30.7	31.9			20.4	13.5	14.9	12.4	6.5	6.1		4.2	3.3	
Primary deficit that stabilizes the debt-to-GDP ratio	4.9	1.3	23.3			47.8	1.8	-0.2	4.8	4.1	3.0		0.2	0.4	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	-0.3	1.9	3.9	2.7	1.7	4.8	5.0	5.2	19.9	19.7	14.3	11.5	2.6	3.1	3.6
Average nominal interest rate on forex debt (in percent)	0.9	1.1	1.3	1.4	0.3	0.9	1.0	1.2	1.4	1.5	1.6	1.3	1.6	1.4	1.5
Average real interest rate on domestic debt (in percent)	5.4	-9.6	-12.6	-3.2	7.5	-4.8	0.2	0.5	2.4	1.8	-0.6	-0.1	-0.4	-1.6	-0.4
Real exchange rate depreciation (in percent, + indicates depreciation)	-9.7	3.9	-2.3	1.0	19.6	-9.3	...	...	...	...	...	…	...	...	...
Inflation rate (GDP deflator, in percent)	6.8	20.2	19.6	17.2	10.3	14.7	6.3	7.4	5.2	5.0	7.4	7.7	4.8	5.3	5.0
Growth of real primary spending (deflated by GDP deflator, in percent)	0.5	0.3	-0.3	0.1	0.2	0.4	0.0	0.0	0.1	0.1	0.1	0.1	0.0	0.1	0.0
Grant element of new external borrowing (in percent)	...	...	...	…	…	39.7	39.7	39.7	39.7	39.7	39.7	39.7	33.4	30.3	...

Sources: Country authorities; and staff estimates and projections.
1/ Public sector covers central government; net debt is used.
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
3/ Revenues excluding grants.
4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4.Guinea: Sensitivity Analysis for Key Indicators of Public Debt 2012-2032

	Projections							
	2012	2013	2014	2015	2016	2017	2022	2032
PV of Debt-to-GDP Ratio								
Baseline	29	29	28	24	20	18	17	10
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	29	30	32	33	33	33	38	45
A2. Primary balance is unchanged from 2012	29	30	33	30	28	27	36	56
A3. Permanently lower GDP growth 1/	29	29	29	24	21	18	20	21
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014	29	30	33	28	25	23	27	30
B2. Primary balance is at historical average minus one standard deviations in 2013-2014	29	32	36	30	26	23	21	13
B3. Combination of B1-B2 using one half standard deviation shocks	29	31	35	30	26	24	27	27
B4. One-time 30 percent real depreciation in 2013	29	35	34	28	24	21	19	13
B5. 10 percent of GDP increase in other debt-creating flows in 2013	29	36	35	30	25	22	21	13
PV of Debt-to-Revenue Ratio 2/								
Baseline	126	120	114	99	92	90	84	39
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	126	124	126	131	149	165	190	178
A2. Primary balance is unchanged from 2012	126	127	132	126	130	139	180	222
A3. Permanently lower GDP growth 1/	126	121	115	101	95	95	98	82
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014	126	126	129	117	114	118	135	120
B2. Primary balance is at historical average minus one standard deviations in 2013-2014	126	134	145	126	118	116	106	53
B3. Combination of B1-B2 using one half standard deviation shocks	126	130	139	125	121	123	134	109
B4. One-time 30 percent real depreciation in 2013	126	145	138	118	109	106	96	53
B5. 10 percent of GDP increase in other debt-creating flows in 2013	126	148	143	123	115	113	104	52
Debt Service-to-Revenue Ratio 2/								
Baseline	17	11	12	11	6	6	4	3
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	17	12	13	13	9	10	8	10
A2. Primary balance is unchanged from 2012	17	11	13	11	7	7	6	10
A3. Permanently lower GDP growth 1/	17	11	12	11	6	6	4	5
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014	17	12	13	12	7	7	5	7
B2. Primary balance is at historical average minus one standard deviations in 2013-2014	17	11	13	12	7	6	5	4
B3. Combination of B1-B2 using one half standard deviation shocks	17	12	13	12	7	7	5	6
B4. One-time 30 percent real depreciation in 2013	17	13	15	14	7	7	5	5
B5. 10 percent of GDP increase in other debt-creating flows in 2013	17	11	13	12	7	6	5	4

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.